

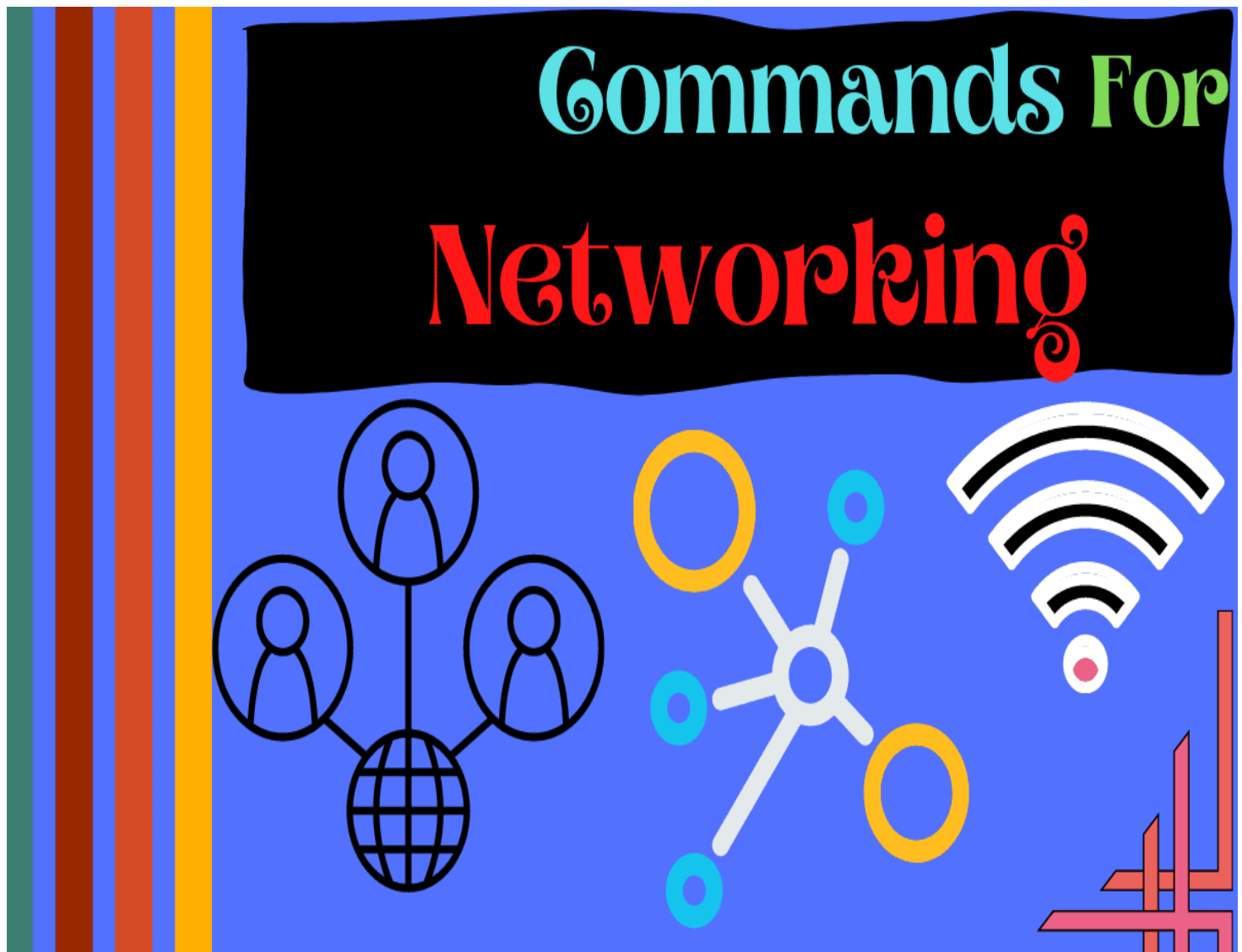


S.P.M College, Udantpuri

Bachelor Of Computer Application (BCA)

Part -2 (Paper-4)

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❖ Networking Command

A command in Networking is an instruction given to the computer or device to perform a specific network related task, such as checking connections, testing communication, or viewing configuration details etc. Example :- ping, ipconfig, tracert, netstat, nslookup,.....

❖ Top Command use in Networking

1. ipconfig (internet protocol Configuration)

- Used to view the IP address, Subnet Mask, MAC address & gateway on the computer or other device/server.
- Windows User
 - ✓ c:\> ipconfig (See ip address)
 - ✓ c:\> ipconfig /all (see all)
- Linux/macOS
 - ✓ C:\> ifconfig

```
C:\Users>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 1:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 10:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix  . : domain.name
    IPv6 Address. . . . . : 2401:4900:1ca9:ddaf:eb61:c5cb:b30c:64d
    Temporary IPv6 Address. . . . . : 2401:4900:1ca9:ddaf:f8e4:1ae1:540:8dc6
    Link-local IPv6 Address . . . . . : fe80::7c86:daa8:6c78:d85f%4
    IPv4 Address. . . . . : 192.168.1.7
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : fe80::6e4f:89ff:fee4:8e29%4
                                192.168.1.1

Ethernet adapter Bluetooth Network Connection:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :
```

2. getmac

- Used to view the MAC address of network device.
- Example → c:\> getmac

```
C:\>getmac

Physical Address      Transport Name
=====
48-BA-AB-EF-23      Media disconnected
D4-6A-1E-46-0F      \Device\Tcpip_{29C9D35F-23BE-4EE6-814B-A841CFB806A6}
D4-6A-1E-46-10      Media disconnected
```

3. hostname

- Used to display the current name of your windows computer.
- Example → c:\> hostname

```
C:\>hostname  
DESKTOP-NDUQFAU
```

4. ping (packet internet groper)

- Used to test network connectivity and measure Round-trip time (RTT) to a destination host.
- Used → Testing b/w two device network connectivity
- Measured → millisecond(ms)
- Protocol → ICMP (internet control message protocol) {main work : check network error, status and diagnostic information, not work of webpage/file.}
- If lower ping → faster connection
- Check two node (LAN/WAN level)
- Example →

C:\> ping website_name / IP address

```
C:\>ping google.com  
Ping request could not find host google.com. Please check the name and try again.
```

(when internet off)

```
C:\>ping rtps.com  
  
Pinging rtps.com [64:ff9b::4cdf:3692] with 32 bytes of data:  
Reply from 64:ff9b::4cdf:3692: time=93ms  
Reply from 64:ff9b::4cdf:3692: time=94ms  
Reply from 64:ff9b::4cdf:3692: time=90ms  
Reply from 64:ff9b::4cdf:3692: time=79ms
```

```
Ping statistics for 64:ff9b::4cdf:3692:  
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),  
Approximate round trip times in milli-seconds:  
Minimum = 79ms, Maximum = 94ms, Average = 89ms
```

```
C:\>ping 8.8.8.8
```

```
Pinging 8.8.8.8 with 32 bytes of data:  
Reply from 8.8.8.8: bytes=32 time=105ms TTL=113  
Reply from 8.8.8.8: bytes=32 time=64ms TTL=113  
Reply from 8.8.8.8: bytes=32 time=218ms TTL=113  
Reply from 8.8.8.8: bytes=32 time=57ms TTL=113
```

```
Ping statistics for 8.8.8.8:  
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),  
Approximate round trip times in milli-seconds:  
Minimum = 57ms, Maximum = 218ms, Average = 111ms
```

```
C:\>
```

- **TTL (Time to Live)** is a field in the IP packet header that prevents a packet from continuously circulating in a **routing loop**.
- ✓ Each time a packet passes through a **router (hop)**, its TTL value is

decreased by **1**.

- ✓ When the TTL reaches **0**, the router discards the packet.
- ✓ TTL therefore limits the **maximum number of hops** a packet can make.
- ✓ TTL can also help determine the number of **routers/hops** between the source and destination, as used by tools such as tracer.
- **Example:**
If a packet starts with TTL = 5, it can pass through up to about **5 router hops** before its TTL expires.
- By default TTL start 64,128,255,....
- **RTT (Round Trip Time)** is the total time taken by a packet to travel from the **source device (PC) to the destination device (server)** and back to the source device.

In the ping command:

- **Minimum RTT** → The shortest time taken by any packet for the round trip.
- **Maximum RTT** → The longest time taken by any packet for the round trip.
- **Average RTT** → The average round-trip time of all the packets.
- **Example:**
 - ✓ If 4 packets have the following RTT values:
 - ✓ 57 ms, 100 ms, 109 ms, 218 ms
 - ✓ Then:
 - ✓ **Minimum = 57 ms**
 - Maximum = 218 ms**
 - Average = $(57 + 100 + 109 + 218) \div 4 = 121$ ms**
- In windows OS Ping command send by default 4 packet, if you want more than 4 then type command prompt :
c:\> ping -n 10 google.com (here 10 packet send)

```

C:\>ping google.com

Pinging google.com [2404:6800:4002:81e::200e] with 32 bytes of data:
Reply from 2404:6800:4002:81e::200e: time=93ms
Reply from 2404:6800:4002:81e::200e: time=70ms
Reply from 2404:6800:4002:81e::200e: time=92ms
Reply from 2404:6800:4002:81e::200e: time=75ms

Ping statistics for 2404:6800:4002:81e::200e:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 70ms, Maximum = 93ms, Average = 82ms

C:\>ping -4 google.com

Pinging google.com [142.250.67.78] with 32 bytes of data:
Reply from 142.250.67.78: bytes=32 time=72ms TTL=113
Reply from 142.250.67.78: bytes=32 time=74ms TTL=113
Reply from 142.250.67.78: bytes=32 time=161ms TTL=113
Reply from 142.250.67.78: bytes=32 time=126ms TTL=113

Ping statistics for 142.250.67.78:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 72ms, Maximum = 161ms, Average = 108ms

C:\>ping -4 rtps.com

Pinging rtps.com [76.223.54.146] with 32 bytes of data:
Reply from 76.223.54.146: bytes=32 time=141ms TTL=240
Reply from 76.223.54.146: bytes=32 time=90ms TTL=240
Reply from 76.223.54.146: bytes=32 time=86ms TTL=240
Reply from 76.223.54.146: bytes=32 time=87ms TTL=240

Ping statistics for 76.223.54.146:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 86ms, Maximum = 141ms, Average = 101ms

C:\>

```

5. nslookup (Name server lookup)

- Find out the corresponding IP address of domain name system(DNS)
- Example :

C:\> nslookup website_name

```

C:\>nslookup rtps.com
Server:  UnKnown
Address:  10.124.86.17

Non-authoritative answer:
Name:     rtps.com
Addresses: 64:ff9b::df8:a930
           64:ff9b::4cdf:3692
           76.223.54.146
           13.248.169.48

C:\>nslookup facebook.com
Server:  UnKnown
Address:  10.124.86.17

Non-authoritative answer:
Name:     facebook.com
Addresses: 2a03:2880:f16b:183:face:b00c:0:25de
           31.13.64.35

```

6. tracert (Trace root)

- This command is used for displaying possible routes(path) and measuring transit delays of packets.
- Protocol – ICMP
- Max – 30 router/HOP

Example

Windows

```
C:\> tracert website_name
```

Linux / macOS

```
C:\> traceroute website_name
```

```
C:\>tracert rtps.com

Tracing route to rtps.com [64:ff9b::4cdf:3692]
over a maximum of 30 hops:

  0  109 ms    *          56 ms    2409:40e4:6d:3ef0::86
  1  46 ms     36 ms     115 ms   2405:200:5213:21:3924:0:3:101
  2  42 ms     36 ms     52 ms    2405:200:5213:21:3925::1
  3  39 ms     28 ms     98 ms    2405:200:858:3631:61::a
  4  45 ms     55 ms     38 ms    64:ff9b::ac11:bd83
  5  42 ms     40 ms     30 ms    64:ff9b::c0a8:8cba
  6  *         *         *        Request timed out.
  7  *         *         *        Request timed out.
  8  *         *         *        Request timed out.
  9  *         *         *        Request timed out.
 10  *         *         *        Request timed out.
 11  *         *         *        Request timed out.
 12  76 ms     78 ms     64 ms    64:ff9b::312c:1226
 13  *         *         *        Request timed out.
 14  *         *         *        Request timed out.
 15  *         *         *        Request timed out.
 16  *         *         *        Request timed out.
 17  *         *         *        Request timed out.
 18  *         *         *        Request timed out.
 19  108 ms    72 ms     64 ms    64:ff9b::345f:4339
 20  67 ms     77 ms     73 ms    64:ff9b::345f:43a6
 21  64 ms     55 ms     61 ms    a904c694c05102f30.awsglobalaccelerator.com [64:ff9b::4cdf:3692]

Trace complete.
```

7.netstat (Network statics)

- Used for check my computer(user) active network connections/state, ports, and protocols (TCP/UDP)
- Example

```
C:\> netstat or
```

```
C:\> netstat -ano
```

```
C:\Users\hira>netstat -ano
```

Active Connections

Proto	Local Address	Foreign Address	State	PID
TCP	0.0.0.0:135	0.0.0.0:0	LISTENING	512
TCP	0.0.0.0:445	0.0.0.0:0	LISTENING	4
TCP	0.0.0.0:2968	0.0.0.0:0	LISTENING	11204
TCP	0.0.0.0:3306	0.0.0.0:0	LISTENING	5540
TCP	0.0.0.0:3389	0.0.0.0:0	LISTENING	1172
TCP	0.0.0.0:5040	0.0.0.0:0	LISTENING	5744
TCP	0.0.0.0:33060	0.0.0.0:0	LISTENING	5540
TCP	0.0.0.0:49664	0.0.0.0:0	LISTENING	856
TCP	0.0.0.0:49665	0.0.0.0:0	LISTENING	776
TCP	0.0.0.0:49666	0.0.0.0:0	LISTENING	1608
TCP	0.0.0.0:49667	0.0.0.0:0	LISTENING	2052
TCP	0.0.0.0:49668	0.0.0.0:0	LISTENING	3016
TCP	0.0.0.0:49669	0.0.0.0:0	LISTENING	3264
TCP	0.0.0.0:49672	0.0.0.0:0	LISTENING	4280
TCP	0.0.0.0:49686	0.0.0.0:0	LISTENING	848
TCP	127.0.0.1:49681	127.0.0.1:49682	ESTABLISHED	5540
TCP	127.0.0.1:49682	127.0.0.1:49681	ESTABLISHED	5540
TCP	127.0.0.1:49683	127.0.0.1:49684	ESTABLISHED	5540
TCP	127.0.0.1:49684	127.0.0.1:49683	ESTABLISHED	5540
TCP	127.0.0.1:54634	127.0.0.1:54635	ESTABLISHED	12032
TCP	127.0.0.1:54635	127.0.0.1:54634	ESTABLISHED	12032
TCP	127.0.0.1:54639	127.0.0.1:54640	ESTABLISHED	17440
TCP	127.0.0.1:54640	127.0.0.1:54639	ESTABLISHED	17440
TCP	192.168.1.9:139	0.0.0.0:0	LISTENING	4
TCP	192.168.1.9:55930	34.107.243.93:443	ESTABLISHED	12032
TCP	192.168.1.9:55942	52.184.215.111:443	CLOSE_WAIT	2548

- Proto → Protocol (TCP/UDP)
- Local Address → Ip and port your PC
- Foreign address → connect which server or also your pc
- State → state of connection (Listing → for connection ready, Established → Connection successfully, close time → server request for connection close)
- PID (Process identifier) : every running program in background window OS provide unique number it is called PID.
Which PID which program you can see → task manager → details

8. ARP (Address Resolution protocol)

- Find MAC address of destination computer.
- Example –

```
C:\>arp -a
```

```
C:\>arp -a

Interface: 10.124.86.27 --- 0x4
    Internet Address      Physical Address      Type
    10.124.86.17          b2-e1-0c-69-2c-40    dynamic
    10.124.86.255         ff-ff-ff-ff-ff-ff    static
    224.0.0.2             01-00-5e-00-00-02    static
    224.0.0.22            01-00-5e-00-00-16    static
    224.0.0.251           01-00-5e-00-00-fb    static
    224.0.0.252           01-00-5e-00-00-fc    static
    255.255.255.255       ff-ff-ff-ff-ff-ff    static
```

9. Pathping

- pathping combines the functions of ping and tracert: it shows the route to a destination and collects packet-loss/latency statistics for the hops along the path.
- This command you can see ISP ip address.

```
C:\Users\hira>pathping -4 google.com

Tracing route to google.com [142.251.221.142]
over a maximum of 30 hops:
  0  hira [192.168.1.9]
  1  192.168.1.1
  2  10.240.12.100
  3  *           *           *
Computing statistics for 50 seconds...
Hop  RTT      Source to Here   This Node/Link   Address
    RTT      Lost/Sent = Pct  Lost/Sent = Pct  Lost/Sent = Pct
  0  0ms       0/ 100 = 0%      0/ 100 = 0%      0/ 100 = 0%      hira [192.168.1.9]
  1  31ms      0/ 100 = 0%      0/ 100 = 0%      0/ 100 = 0%      192.168.1.1
  2  54ms      0/ 100 = 0%      0/ 100 = 0%      0/ 100 = 0%      10.240.12.100
Trace complete.
```

10. netsh (network shell)

- netsh is a command-line tool in Windows used to view, configure, modify, and troubleshoot network settings.

Command	Use
netsh wlan show interfaces	Shows information about the current Wi-Fi connection

Command

netsh wlan show profiles

netsh wlan show drivers

netsh interface ip show config

netsh wlan show profile name="wifi name"
key=clear

Example : C:\>netsh wlan show profile
name="airtel_black&white" key=clear

netsh winsock reset

Use

Shows saved Wi-Fi network profiles

Shows Wi-Fi adapter and driver information

Shows IP configuration of network interfaces

Shows password current wifi

Resets the Windows Winsock configuration

```
C:\Users\hira>netsh
netsh>wlan show interfaces

There is 1 interface on the system:

Name                : Wi-Fi
Description          : Realtek RTL8723DE 802.11b/g/n PCIe Adapter
GUID                 : 29c9d35f-23be-4ee6-814b-a841cfb806a6
Physical address     : d4:6a:6a:1e:46:0f
State                : connected
SSID                 : Airtel_Black & White
BSSID                : 6c:4f:89:e4:8e:2f
Network type         : Infrastructure
Radio type           : 802.11n
Authentication       : WPA2-Personal
Cipher               : CCMP
Connection mode      : Auto Connect
Channel              : 11
Receive rate (Mbps)  : 57.8
Transmit rate (Mbps) : 57.8
Signal               : 31%
Profile              : Airtel_Black & White

Hosted network status : Not available

netsh>netsh wlan show networks
The following command was not found: netsh wlan show networks.
netsh>wlan show networks

Interface name : Wi-Fi
There are 20 networks currently visible.

SSID 1 : Airtel_Black & White
Network type           : Infrastructure
```

```

netsh>wlan show profile name="airtel_black & white" key=clear

Profile Airtel_Black & White on interface Wi-Fi:
=====

Applied: All User Profile

Profile information
-----
Version           : 1
Type              : Wireless LAN
Name              : Airtel_Black & White
Control options   :
    Connection mode : Connect automatically
    Network broadcast : Connect only if this network is broadcasting
    AutoSwitch       : Do not switch to other networks
    MAC Randomization : Disabled

Connectivity settings
-----
Number of SSIDs    : 1
SSID name          : "Airtel_Black & White"
Network type       : Infrastructure
Radio type         : [ Any Radio Type ]
Vendor extension    : Not present

Security settings
-----
Authentication     : WPA2-Personal
Cipher              : CCMP
Authentication     : WPA2-Personal
Cipher              : GCMP
Security key        : Present
Key Content         : 88311110

Cost settings
-----

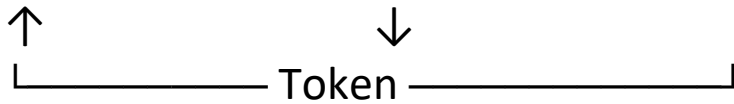
```

Command End

❖ Token Passing

- Token Passing is a MAC (Media Access Control) protocol in which a special control frame called a token circulates among the devices on a network. A device can transmit data only when it possesses the token.
- Example: Token Ring network.
- Working:

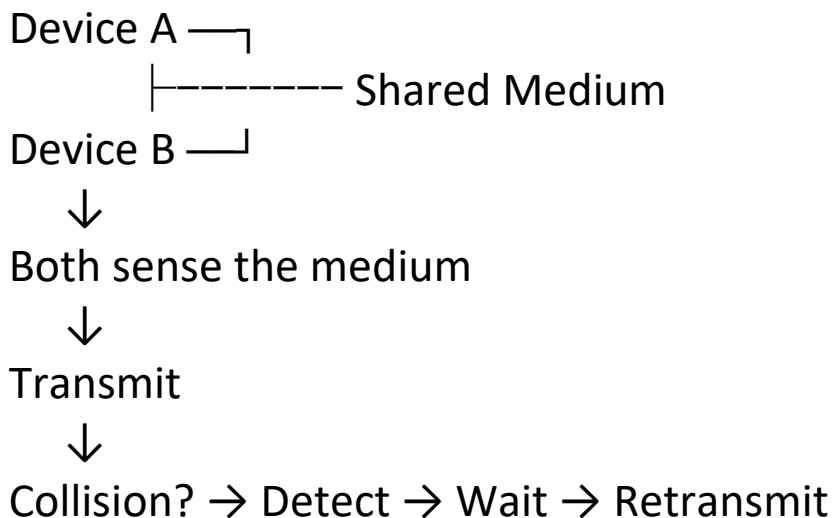
Computer A → Computer B → Computer C → Computer D



- Only the computer holding the token can transmit, so **collisions are avoided**.

❖ CSMA/CD protocol

- CSMA/CD (Carrier Sense Multiple Access with Collision Detection) is a MAC protocol in which a device listens to the shared medium before transmitting. If the medium is free, it transmits; if a collision occurs, the devices detect it, stop transmission, wait for a random time, and retransmit.
- Mainly associated with: Traditional half-duplex Ethernet.
- Working:



❖ Difference b/w token passing & CSMA/CD

Feature	Token Passing	CSMA/CD
Basic idea	Device must have the token to transmit	Device senses the medium before transmitting
Collision	No collision in normal operation	Collision can occur
Access method	Controlled access	Contention-based access
Waiting	Waits for the token	Waits if the medium is busy or after collision
Collision handling	Not normally required	Detects and handles collisions

Predictability	More predictable	Less predictable under heavy traffic
Example	Token Ring	Traditional Ethernet

❖ Server

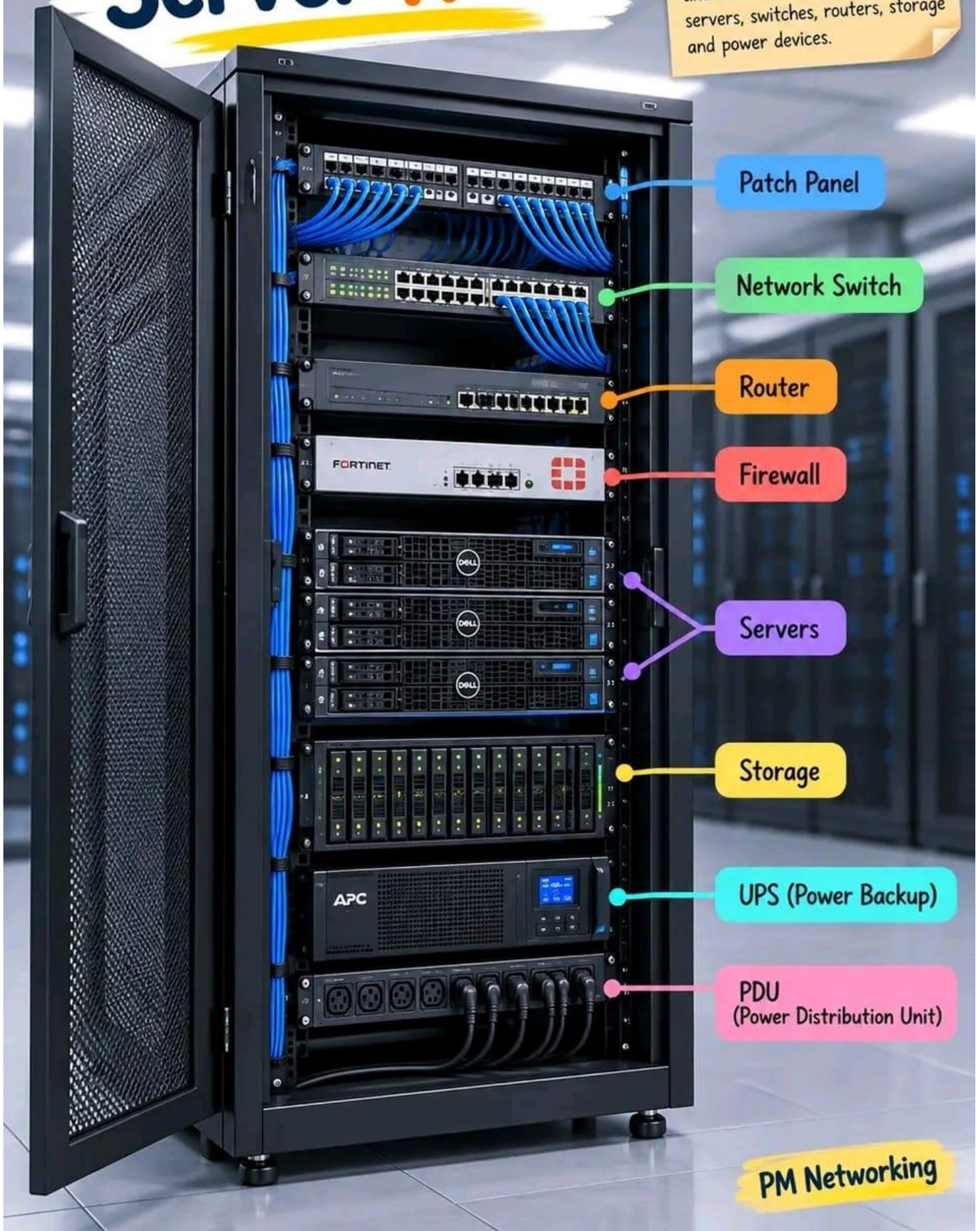
- A server is a computer/system that receives requests from clients (your computer) and provides the requested service or data.
- The **client** requests a webpage, and the **server** sends the webpage/data back.
- **Server does not always mean a special physical computer.**
- A **server can be**:
 - A physical computer
 - A virtual machine
 - A software application running on a computer

❖ Types of Server

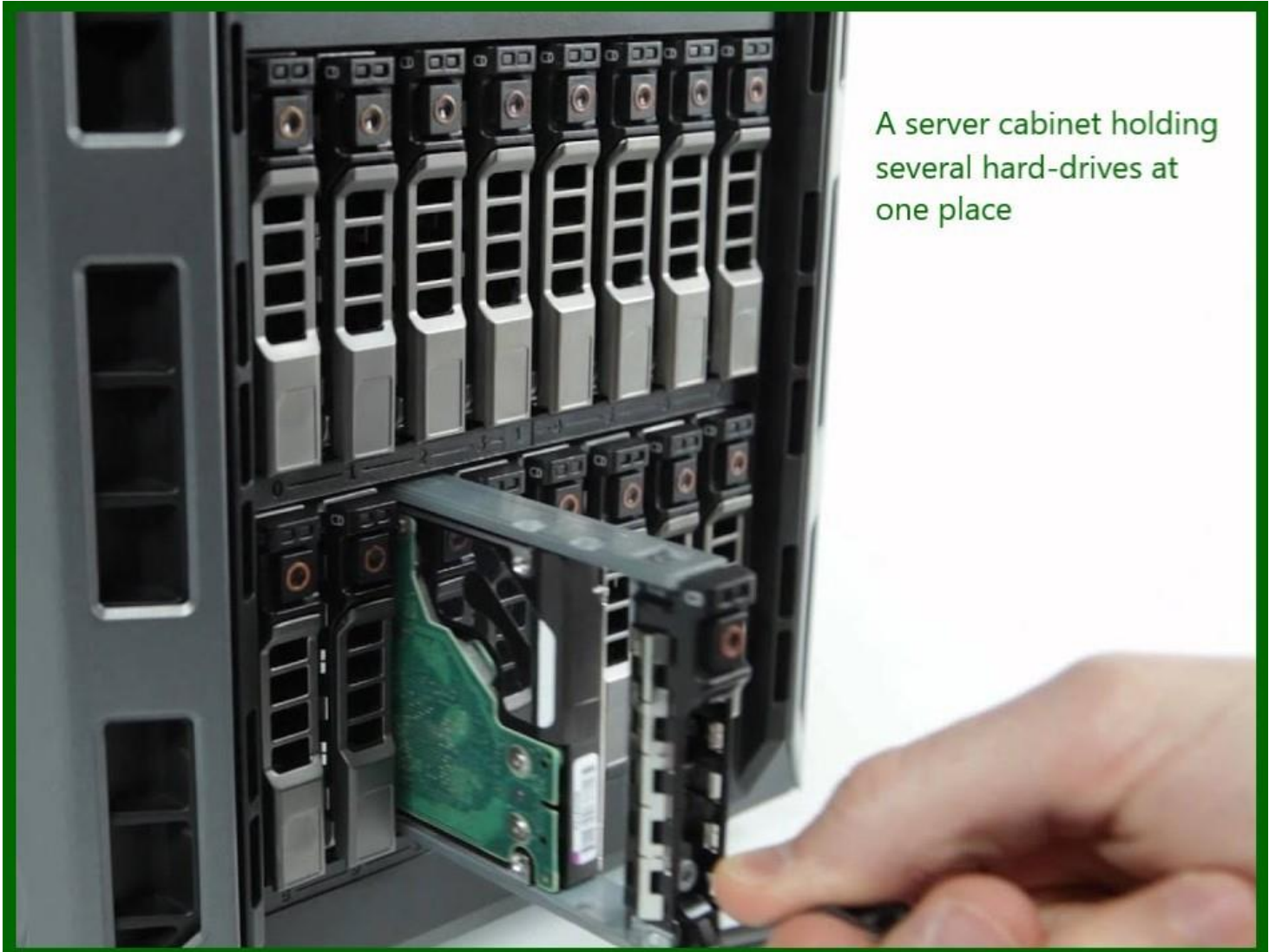
Server	Main Function
Web Server	Provides web pages/web content
File Server	Stores and provides files
Mail Server	Sends, receives, and stores emails
DNS Server	Resolves domain names into IP addresses
Database Server	Stores and provides database information
DHCP Server	Automatically assigns IP configuration
Proxy Server	Acts as an intermediary between client and Internet

Server Rack

A server rack is a standardized cabinet used to organize, secure and house IT equipment like servers, switches, routers, storage and power devices.

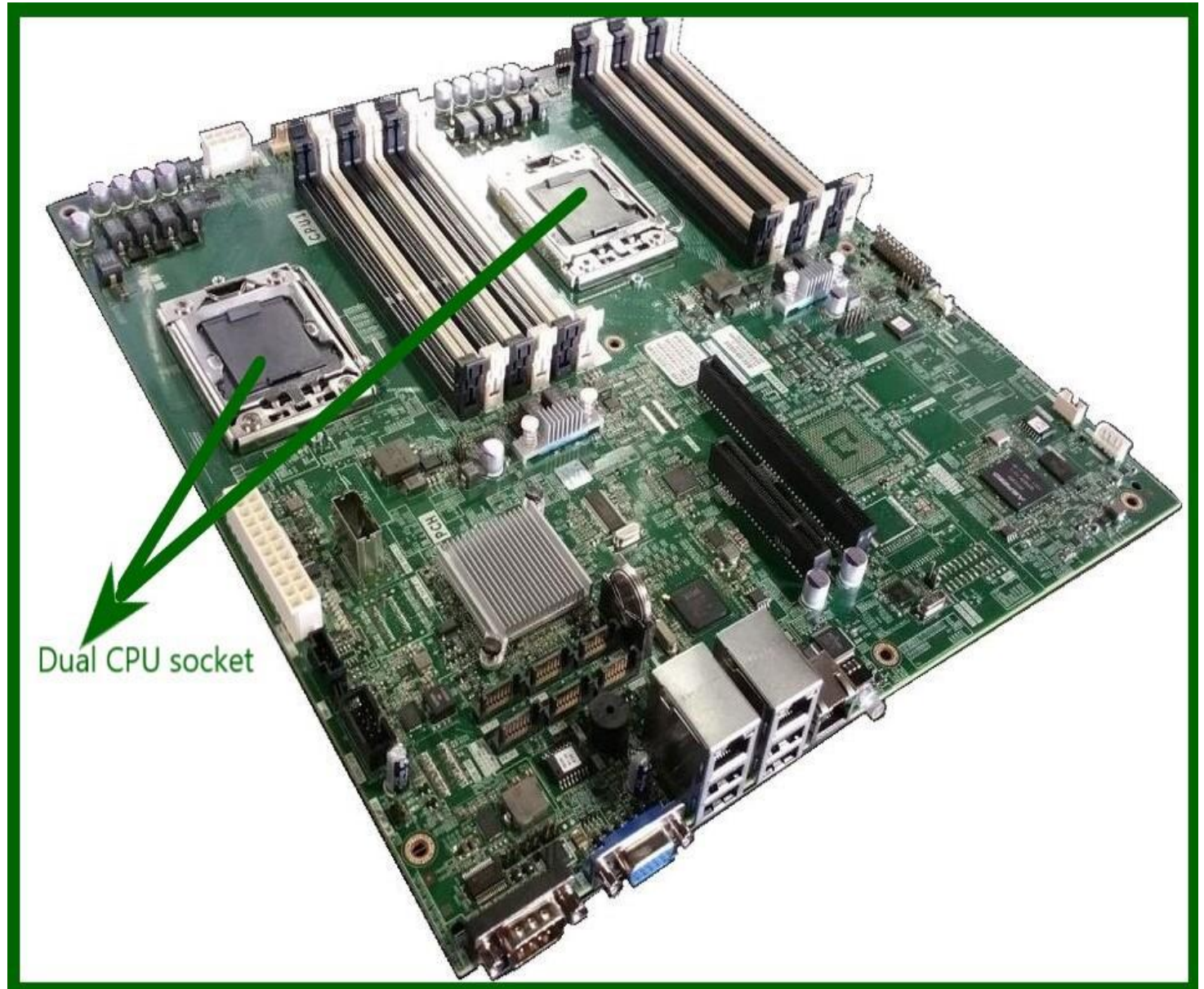


❖ Server computer



A server cabinet holding
several hard-drives at
one place

The motherboard also is expensive and comes with a dual or quad or even octa- CPU configuration and supports large amount of RAM (up to 1-2TB of Ram). Also, server motherboard can support many hard drives.



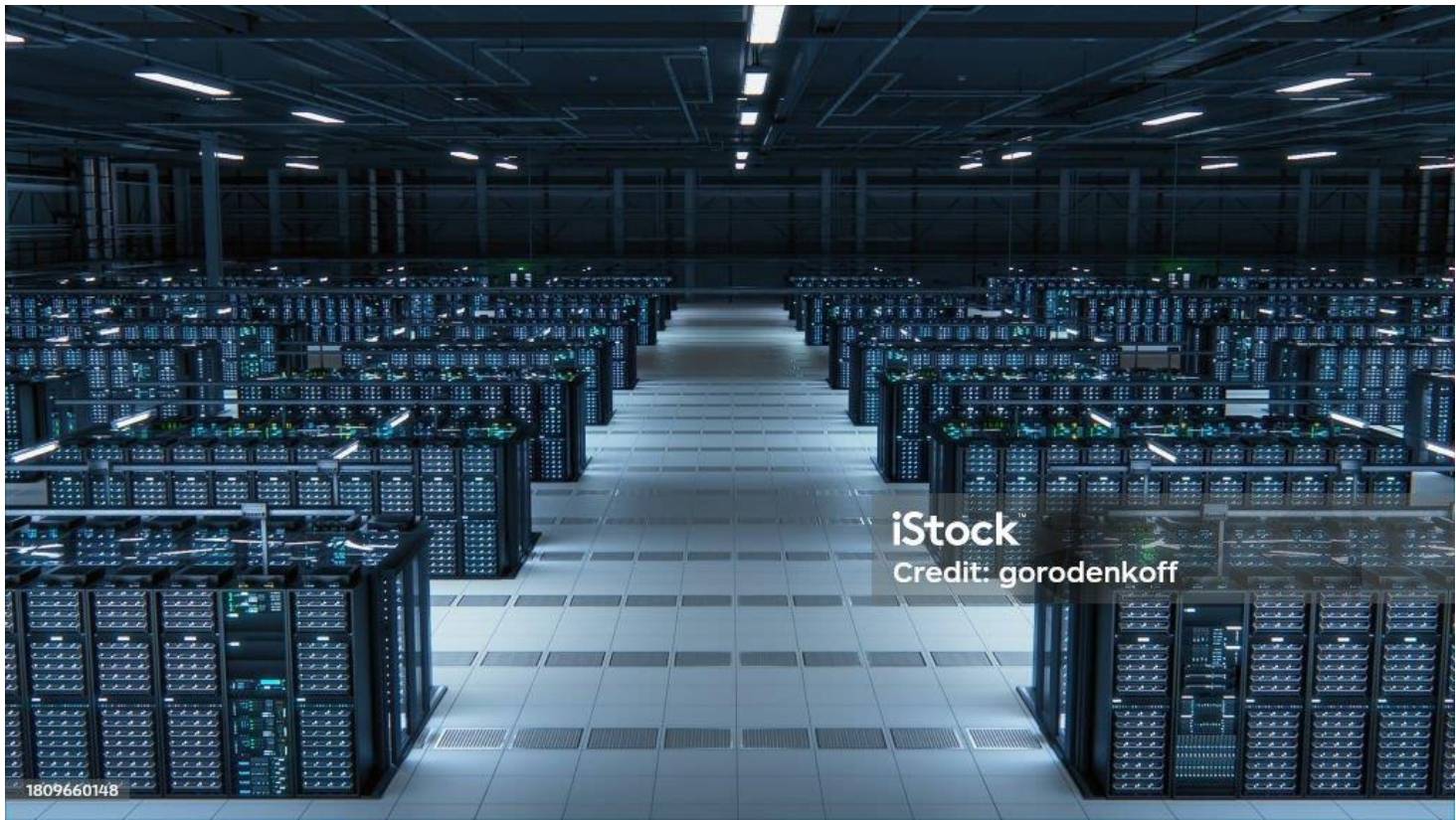
❖ Data center

A data center is a facility that houses computer servers and networking equipment to store, process, manage, and deliver data and services.

(भारत में बहुत बड़ी संख्या में data centers हैं। इन्हें broadly दो तरह से समझिए:)

1. **Commercial Data Centers** — Sify, CtrlS, NTT, Yotta, Nxtra, STT GDC,...
2. **Cloud/Technology companies के Data Centers** — AWS, Microsoft Azure, Google Cloud, Oracle,.....
3. **Government Data Centers** — NIC National Data Centres और State Data Centres।

State/UT	प्रमुख शहर	कुछ प्रमुख companies/data-center operators
Maharashtra	Mumbai, Navi Mumbai, Pune, Nashik	AWS, Microsoft, Google, NTT, Sify, CtrlS, Yotta, Web Werks, Jio, ESDS
Telangana	Hyderabad	AWS, CtrlS, Oracle, Microsoft, TCS, Sify
Karnataka	Bengaluru	NTT, Sify, CtrlS, Nxtra, ITI, ESDS
Tamil Nadu	Chennai	Microsoft, IBM, AWS ecosystem, Sify, STT GDC
Uttar Pradesh	Noida, Greater Noida	Google, Nxtra, CtrlS, Sify, Web Werks, Yotta
West Bengal	Kolkata	Sify, CtrlS
Gujarat	Ahmedabad, Jamnagar	BSNL, Jio, अन्य operators
Rajasthan	Jaipur	Cyfuture
Andhra Pradesh	Guntur	Pi Datacenters, Yotta (DR)
Haryana	Gurugram/Faridabad	RailTel, BSNL, अन्य operators
Delhi	New Delhi	Tata Communications, NIC/RailTel आदि
Punjab/Chandigarh	Chandigarh	Sify
Kerala	Kochi	Sify
Bihar	Patna	Government/State infrastructure और विभिन्न network/IT facilities



DATA CENTER COOLING

01



AIR-BASED COOLING

Uses CRAC/CRAH units, chilled air, and hot/cold aisle containment to regulate server temperatures.



02



LIQUID COOLING

Circulates coolant directly to servers or components through cold plates or Liquid-cooled racks for high heat removal.



03



IMMERSION COOLING

Servers are submerged in a non-conductive dielectric fluid that absorbs and dissipates heat efficiently.



04



EVAPORATIVE COOLING

Utilizes water evaporation to cool incoming air significantly reducing energy consumption in suitable climates.



05



FREE COOLING ECONOMIZER COOLING!

Uses water-cooled units, chilled air, to room to past tiny cold air hot cold retrouard from enagr, refrigeration, reduucing energy efficiency



Reduced Mechanical Cooling